

Discovering Inverse Functions

A Visual Exploration

Name: _____

Date: _____

Period: _____

Learning Target: I can discover the relationship between a function and its inverse by swapping x and y values and observing the graphical result.

Example 1: The Parent Function $f(x) = \sqrt{x}$

Step 1: Complete the table for $f(x) = \sqrt{x}$. **Step 2:** Swap the x and y values to create the inverse table.

Original Function

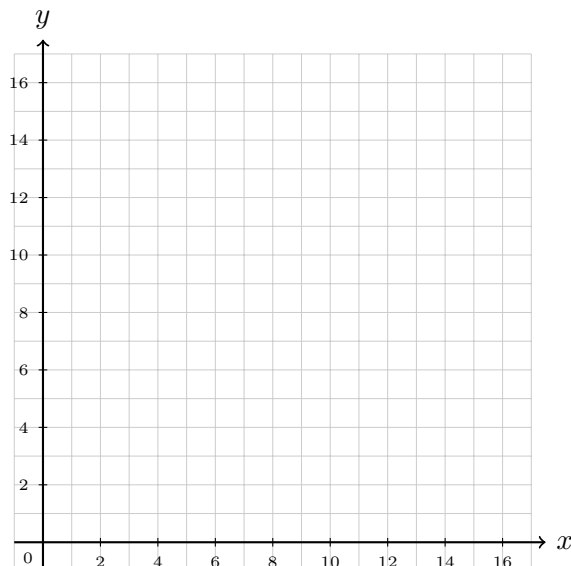
$f(x) = \sqrt{x}$

x	y
0	
1	
4	
9	
16	

Inverse (Swapped)

$f^{-1}(x) = \underline{\hspace{2cm}}$

x	y

Step 3: Graph both sets of points. Draw $y = x$ as a dashed line.

What Do You Notice?

1. What is the relationship between the two curves and the line $y = x$?

2. If the original function passes through $(4, 2)$, what point does the inverse pass through?

3. What equation describes the inverse curve? $f^{-1}(x) = \underline{\hspace{2cm}}$

Example 2: A Transformed Function $f(x) = \sqrt{x-2} + 1$ **Step 1:** What is the endpoint of this function? (____, ____)**Step 2:** Complete both tables, then graph.**Original Function**

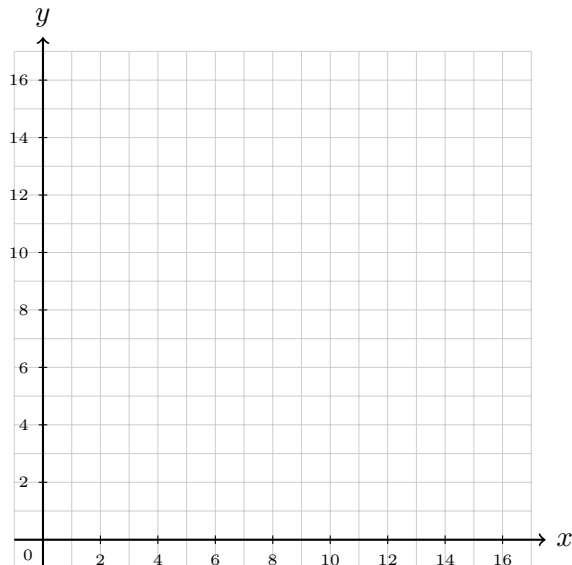
$f(x) = \sqrt{x-2} + 1$

x	y
2	
3	
6	
11	
18	

Inverse (Swapped)

$f^{-1}(x) = \underline{\hspace{2cm}}$

x	y

Step 3: Graph both. Draw $y = x$ as a dashed line.**Step 4:** What is the endpoint of the inverse function? (____, ____)**Step 5:** How are the endpoints of $f(x)$ and $f^{-1}(x)$ related? _____**KEY CONCEPT: Inverse Functions**

- When you **swap the x and y values** of a function, you get its **inverse function**.
- The graphs of $f(x)$ and $f^{-1}(x)$ are **reflections over the line $y = x$** .
- If $f(a) = b$, then $f^{-1}(b) = a$. (The input and output switch roles!)
- The **endpoint of the inverse** is the endpoint of the original with coordinates swapped.

Practice: Finding the Inverse Algebraically

- To find the inverse of a square root function:
- (1) Replace $f(x)$ with y
- (2) Swap x and y
- (3) Solve for y (isolate the radical, then square both sides)
- (4) Write as $f^{-1}(x)$

Example: Find the inverse of $f(x) = \sqrt{x-2} + 1$

- $y = \sqrt{x-2} + 1$
- $x = \sqrt{y-2} + 1$
- $x - 1 = \sqrt{y-2}$
- $(x - 1)^2 = y - 2$
- $f^{-1}(x) = (x - 1)^2 + 2$

Your Turn: Find the inverse of each function. Then complete the tables and graph.

Problem 1: $f(x) = \sqrt{x+4} - 2$

Show your algebraic work to find $f^{-1}(x)$:

$f^{-1}(x) =$ _____

Endpoint of $f^{-1}(x)$: (_____, _____)

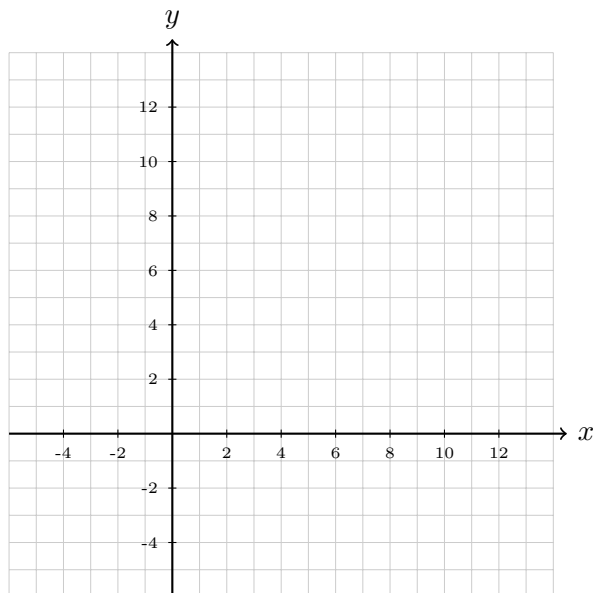
Original Function

Inverse (Swapped)

x	y
-4	
-3	
0	
5	
12	

x	y

Graph



Problem 2: $f(x) = \sqrt{x-4} + 3$

Show your algebraic work to find $f^{-1}(x)$:

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

Endpoint of $f^{-1}(x)$: (____, ____)

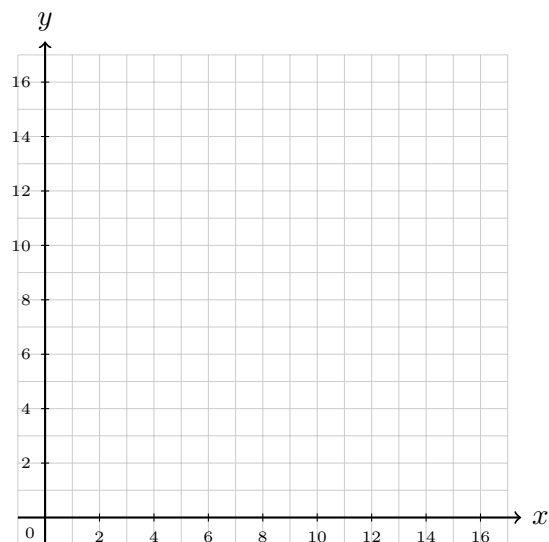
Original Function

Inverse (Swapped)

x	y
4	
5	
8	
13	
20	

x	y

Graph



Problem 3: $f(x) = \sqrt{x-1} + 3$

Show your algebraic work to find $f^{-1}(x)$:

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

Endpoint of $f^{-1}(x)$: (____, ____)

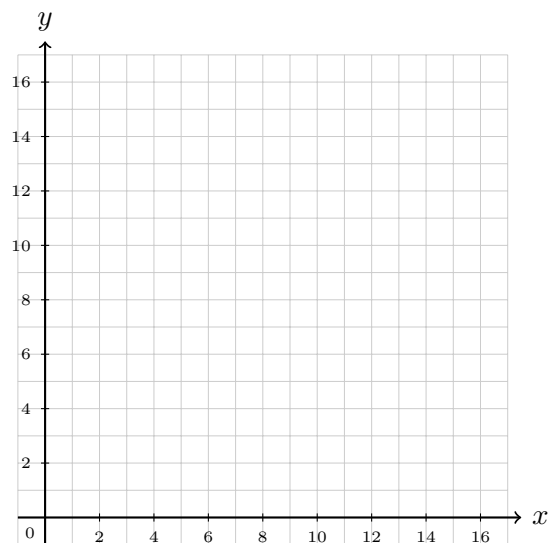
Original Function

Inverse (Swapped)

x	y
1	
2	
5	
10	
17	

x	y

Graph



Exit Ticket

1. If $f(3) = 7$, what is $f^{-1}(7)$? _____
2. The graph of $f(x)$ passes through $(5, 2)$. What point is on the graph of $f^{-1}(x)$? _____
3. The graphs of $f(x)$ and $f^{-1}(x)$ are reflections over what line? _____
4. If $f(x) = \sqrt{x - 5} + 2$ has endpoint $(5, 2)$, what is the endpoint of $f^{-1}(x)$? _____

ANSWER KEY

Example 1: $f(x) = \sqrt{x}$

Original Table: (0, 0), (1, 1), (4, 2), (9, 3), (16, 4)

Swapped Table: (0, 0), (1, 1), (2, 4), (3, 9), (4, 16)

Inverse: $f^{-1}(x) = x^2$ (for $x \geq 0$)

Discovery Questions:

1. The two curves are reflections of each other over the line $y = x$.
2. The inverse passes through (2, 4).
3. $f^{-1}(x) = x^2$

Example 2: $f(x) = \sqrt{x - 2} + 1$

Endpoint of $f(x)$: (2, 1)

Original Table: (2, 1), (3, 2), (6, 3), (11, 4), (18, 5)

Swapped Table: (1, 2), (2, 3), (3, 6), (4, 11), (5, 18)

Endpoint of $f^{-1}(x)$: (1, 2)

Relationship: The endpoints are swapped (coordinates reversed).

Problem 1: $f(x) = \sqrt{x + 4} - 2$

Endpoint of $f(x)$: (-4, -2)

Inverse: $f^{-1}(x) = (x + 2)^2 - 4$

Endpoint of $f^{-1}(x)$: (-2, -4)

Original Table: (-4, -2), (-3, -1), (0, 0), (5, 1), (12, 2)

Swapped Table: (-2, -4), (-1, -3), (0, 0), (1, 5), (2, 12)

Problem 2: $f(x) = \sqrt{x - 4} + 3$

Endpoint of $f(x)$: (4, 3)

Inverse: $f^{-1}(x) = (x - 3)^2 + 4$

Endpoint of $f^{-1}(x)$: (3, 4)

Original Table: (4, 3), (5, 4), (8, 5), (13, 6), (20, 7)

Swapped Table: (3, 4), (4, 5), (5, 8), (6, 13), (7, 20)

Problem 3: $f(x) = \sqrt{x - 1} + 3$

Endpoint of $f(x)$: (1, 3)

Inverse: $f^{-1}(x) = (x - 3)^2 + 1$

Endpoint of $f^{-1}(x)$: (3, 1)

Original Table: (1, 3), (2, 4), (5, 5), (10, 6), (17, 7)

Swapped Table: (3, 1), (4, 2), (5, 5), (6, 10), (7, 17)

Exit Ticket

1. $f^{-1}(7) = 3$
2. (2, 5)

3. $y = x$

4. $(2, 5)$